THE INDIAN INTERNATIONAL SCHOOL,DSO

orksheet – Chapter 5: Continuity and Differentiability (2026-2027)

Grade: 12 Subject: MATHEMATICS

Name: _____

Date: _____

Instructions:

- This worksheet has 20 questions arranged in three levels of increasing difficulty.
- Level 1: Q 1 – Q 7 (MCQs, Assertion–Reason and short answer questions)
- Level 2: Q 8 – Q 14 (short answer questions)
- Level 3: Q 15 – Q 20 (long answer and case-based questions)
- Marks for each question are given against it. All questions are compulsory.
- The answer key is given at the end of the worksheet.

Q No.	Question	Marks
	LEVEL 1 **Q 1 – Q 7**	**8**
1.	The value of k for which $f(x) = \begin{cases} kx + 1, & x \leq 5 \\ 3x - 5, & x > 5 \end{cases}$ is continuous at $x = 5$ is: (A) $\frac{9}{5}$ (B) $\frac{5}{9}$ (C) $\frac{11}{5}$ (D) 2	1
2.	The function $f(x) = \lvert x \rvert$ is: (A) continuous and differentiable everywhere (B) continuous everywhere but not differentiable at $x = 0$ (C) differentiable but not continuous at $x = 0$ (D) neither continuous nor differentiable at $x = 0$	1
3.	The derivative of e^{e^x} with respect to x is: (A) $\log x$ (B) e^{e^x} (C) $e^x \cdot e^{e^x}$ (D) $(e^x)^2$	1
4.	If $y = \log(\cos e^x)$, then $\frac{dy}{dx}$ is: (A) $\tan e^x$ (B) $-e^x \tan e^x$ (C) $e^x \tan e^x$ (D) $-\tan e^x$	1
5.	If $x = at^2$ and $y = 2at$, then $\frac{dy}{dx}$ is: (A) t (B) $\frac{1}{t}$ (C) $2t$ (D) $-\frac{1}{t^2}$	1

Q No.	Question	Marks
6.	**Assertion (A):** The greatest integer function $f(x) = [x]$ is discontinuous at $x = 2$. **Reason (R):** $\lim_{x\to 2^-}[x] \neq \lim_{x\to 2^+}[x]$. (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.	1
7.	Differentiate $\sin^{-1}\left(2x\sqrt{1-x^2}\right)$ with respect to x, where $-\frac{1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}}$.	2
	LEVEL 2 **Q 8 – Q 14**	**18**
8.	Find k so that $f(x) = \begin{cases} \frac{1-\cos 4x}{8x^2}, & x \neq 0 \\ k, & x = 0 \end{cases}$ is continuous at $x = 0$.	2
9.	Find $\frac{dy}{dx}$ if $y = x^{\sin x} + (\sin x)^x$.	3
10.	If $x = a(\theta + \sin\theta)$ and $y = a(1 - \cos\theta)$, find $\frac{dy}{dx}$.	2
11.	If $y = e^{a\cos^{-1}x}$, $-1 < x < 1$, show that $(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} - a^2y = 0$.	3
12.	If $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$, show that $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$.	3
13.	Find $\frac{dy}{dx}$ if $x^2 + 2xy + y^3 = 42$.	2
14.	Find the values of a and b such that $f(x) = \begin{cases} 5, & x \leq 2 \\ ax + b, & 2 < x < 10 \\ 21, & x \geq 10 \end{cases}$ is continuous.	3
	LEVEL 3 **Q 15 – Q 20**	**25**
15.	If $y = (\tan^{-1}x)^2$, show that $(x^2+1)^2\frac{d^2y}{dx^2} + 2x(x^2+1)\frac{dy}{dx} = 2$.	3
16.	If $(x-a)^2 + (y-b)^2 = c^2$ for some $c > 0$, prove that $\frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}}$ is a constant independent of a and b.	5
17.	If $x^y = e^{x-y}$, prove that $\frac{dy}{dx} = \frac{\log x}{(1+\log x)^2}$.	3
18.	Show that $f(x) = \lvert x-1\rvert + \lvert x-2\rvert$ is continuous at $x = 1$ and $x = 2$ but not differentiable at these points.	5
19.	If $x = a\left(\cos\theta + \log\tan\frac{\theta}{2}\right)$ and $y = \sin\theta$, then find $\frac{d^2y}{dx^2}$ at $\theta = \frac{\pi}{4}$.	5

Q No.	Question	Marks
20.	**Case Study:** While designing a smooth ramp, an engineer models its profile by $f(x) = \begin{cases} x^2 + 3x + p, & x \leq 1 \\ qx + 2, & x > 1 \end{cases}$. For a smooth ramp, f must be differentiable at $x = 1$. **(i)** Write the condition on p and q for f to be continuous at $x = 1$. **[1]** **(ii)** Find the left-hand derivative and right-hand derivative of f at $x = 1$. **[1]** **(iii)** Find the values of p and q. **[2]**	4

***** End of Worksheet *****

ANSWER KEY

Chapter 5: Continuity and Differentiability

Q No.	Answer / Key Steps
	LEVEL 1
1.	(A) $5k + 1 = 10 \Rightarrow k = \frac{9}{5}$
2.	(B)
3.	(C) $e^x \cdot e^{e^x}$
4.	(B) $-e^x \tan e^x$
5.	(B) $\frac{2a}{2at} = \frac{1}{t}$
6.	(A) Both true; R explains A.
7.	Put $x = \sin\theta$: $y = 2\sin^{-1}x$, so $\frac{dy}{dx} = \frac{2}{\sqrt{1-x^2}}$
	LEVEL 2
8.	$\lim_{x\to 0} \frac{2\sin^2 x}{8x^2} = 1$, so $k = 1$
9.	$x^{\sin x}\left(\cos x \log x + \frac{\sin x}{x}\right) + (\sin x)^x(\log\sin x + x\cot x)$
10.	$\frac{a\sin\theta}{a(1+\cos\theta)} = \tan\frac{\theta}{2}$
11.	$y_1 = -\frac{ay}{\sqrt{1-x^2}} \Rightarrow \sqrt{1-x^2}\, y_1 = -ay$; differentiate again and multiply by $\sqrt{1-x^2}$: $(1-x^2)y_2 - xy_1 = a^2 y$.
12.	Put $x = \sin A$, $y = \sin B$: $\cot\frac{A-B}{2} = a \Rightarrow A - B = 2\cot^{-1}a$, i.e. $\sin^{-1}x - \sin^{-1}y =$ constant. Differentiate: $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$.
13.	$\frac{dy}{dx} = -\frac{2(x+y)}{2x+3y^2}$
14.	$2a + b = 5$, $10a + b = 21 \Rightarrow a = 2, b = 1$
	LEVEL 3
15.	$y_1 = \frac{2\tan^{-1}x}{1+x^2} \Rightarrow (1+x^2)y_1 = 2\tan^{-1}x$; differentiate and multiply by $(1+x^2)$ to get the result.
16.	$y_1 = -\frac{x-a}{y-b}$, $y_2 = -\frac{c^2}{(y-b)^3}$; the expression $= -c$, a constant.
17.	$y\log x = x - y \Rightarrow y = \frac{x}{1+\log x}$; $\frac{dy}{dx} = \frac{\log x}{(1+\log x)^2}$
18.	$f(x) = 3 - 2x$ $(x < 1)$, 1 $(1 \le x \le 2)$, $2x - 3$ $(x > 2)$. Continuous at both points. At $x = 1$: LHD $= -2$, RHD $= 0$; at $x = 2$: LHD $= 0$, RHD $= 2$. Not differentiable.
19.	$\frac{dx}{d\theta} = \frac{a\cos^2\theta}{\sin\theta}$, $\frac{dy}{dx} = \frac{\tan\theta}{a}$, $\frac{d^2y}{dx^2} = \frac{\sin\theta}{a^2\cos^4\theta}$; at $\theta = \frac{\pi}{4}$: $\frac{2\sqrt{2}}{a^2}$
20.	(i) $4 + p = q + 2$. (ii) LHD $= 2(1) + 3 = 5$, RHD $= q$. (iii) $q = 5, p = 3$